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ORIGINAL PAPER

Waste water management and water quality of river Yamuna in the megacity of Delhi

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Abstract The present study aims at assessing water quality of river Yamuna in one of the world's most polluted and populated megacities, Delhi. Conductivity, salinity and sodium content were within the permissible categories. Chloride concentration exceeded acceptable levels of drinking water guidelines. Water quality was poor at all locations with respect to heavy metal contamination, especially along the lower section of the Delhi stretch. Heavy metal concentrations were manifold higher than the acceptable limits of drinking water according to the BIS guidelines and reached ~29, 4.9, 10, 31, 27, 83, 7.3 and 27 times higher, respectively, for metals aluminum, copper, chromium, cadmium, iron, lead, manganese and nickel. The Najafgarh and the Shahdara drains are major point sources. Low oxidation–reduction potential reflected high organic loads and traces of eutrophication together with significant levels of nitrate and total phosphate. Discharges from agriculture, sewage and power plants could be important sources of high metal concentration. This calls for urgent measures to be taken for prevention of metal contamination in the river, through both, technology as well as implementation of regulations in order to sustain huge populations in megacities like Delhi. Waste water treatment from point sources needs tremendous improvement on the city. Treatment of the entire waste generated up to the tertiary level is required for minimizing dissolved solids, especially toxic metals, and rendering reuse in agriculture suitable. Treatment plants need proper operation, maintenance, uninterrupted power supply and regular monitoring. Various measure and programmes need to be undertaken to ensure safe reuse of wastewater.

Keywords Heavy metals · Hydrochemistry · Major ions nutrients · Pollution · Sewage treatment · Toxic metals · Waste water treatment

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Introduction

The availability of clean water at the required quantity is a major challenge in metropolitan cities which are the centers of exhaustive developmental activities. Various cities around the world have encountered the crisis. Developing countries like India are still dependent on rivers for various activities, including cleansing as well as disposal of wastes. About 1.8 million people, mostly children die every year from diseases related to contaminated water in the developing countries (WHO 2006). It is, therefore, pertinent to study the impact of anthropogenic activities on the quality of river water.

River Yamuna accounts for more than 70% of Delhi's water supplies. Delhi competes strongly to be among the most populated megacities of the world. The Delhi segment of 22 km that the river traverses from the Wazirabad barrage to the Okhla barrage is responsible for ~80% of the pollution load of the river (Parween et al. 2014, 2016). The agricultural belts of Haryana, Uttar Pradesh (UP) and Delhi, as well as the industrial township all along the river discharge significant amounts of wastes into the river, turning it into a sewer, especially during the dry seasons.

Various researchers have carried out the hydro-chemical characterization of natural waters in the basin as well in the urban area. However, heavy metals too are a major concern because of their toxicity, persistence and bioaccumulation. These metals may be geological in origin and enter the river system by weathering and erosion, or anthropogenic in nature arising out of activities such as mining, industrial and agricultural activities, traffic emission, sewage disposal. The metals can be transferred among different compartments or components of the river ecosystem by partitioning among the components. Poisoning with arsenic, cadmium, chromium and lead is quite well known. Therefore, these require close monitoring within natural water systems, such as rivers which are lifelines to the human populations.

Though studies have been conducted to determine contamination of the river Yamuna with heavy metals in the Delhi stretch (Singh 2001; Trisal et al. 2008), they lie scattered with only a few parameters determined in each study. Major ions distributions have been investigated by some researchers (Arora and Mehra 2003; Ravindra and Kaushik 2003), while others investigated the physicochemical parameters of river Yamuna (Mandal et al. 2010; Upadhyay et al. 2011; Dhillon et al. 2013). However, all the three groups of components are important and interrelated. Diverse kinds of contaminants may share a common source. Different aspects were studied by investigators in different time periods, mostly outdated making comparisons difficult. No systematic study has been conducted in the region to study the heavy metal distribution in relation to the nutrients, major ions concentration as well as other physicochemical parameters of the river. Therefore, the present study provides a comprehensive research on the current status of water pollution in river Yamuna with respect to the above mentioned as well as their interrelationship. This investigation is expected to provide an appropriate reflection of the deteriorated status of waste water management in Delhi region and water quality of river Yamuna.

It is basically important to focus on the prevention of pollution from point sources as against the diffuse ones, which are difficult to deal with. Waste water in this regard are the major thrust areas that need immediate attention. There is an urgent need of not only protecting the river stretch from pollution, but also the recycling of treated waste water for preventing overexploitation of the water resource. Despite that, wastewater treatment and reuse remains a widely neglected field in India.

Voluminous wastewater generation from numerous sectors such as industrial, domestic, agricultural is threat to the depleting water resources as well as the quality of river waters. About 70–80% of total water which is supplied for domestic use turns to wastewater according to the Central Public Health and Environmental Engineering Organisation, Govt. of India. Wastewater generated per capita by the class I cities (classified on the basis of population: more than hundred thousand) and class II towns (population size: 50–100 thousand) comprising of 72% of the total urban population in India amounts to approximately 98 l per capita per day (lpcd), with over 220 lpcd alone generated by the National Capital Territory of Delhi, where only 61% of the total 3663 million liters per day (MLD) of wastewaters is treated before being discharged into the river Yamuna (CPCB 1999).

Leaving aside the domestic sewage, around 13,468 MLD of wastewater is produced by industries, out of which only 60% is being treated. Most of the small scale industries cannot afford the cost of treating the waste it generates. Common effluent treatment plants (CETP) are setup for clusters of such small scale industries (CPCB 2005b). There are 10 CETPs in Delhi with a capacity of 133 MLD which discharge their effluents in the river Yamuna.

According to a report of CPCB (2005a, b), the sewage generated in Delhi has been estimated to be 2947.8 MLD, while the installed sewage treatment capacity is reported as 2330.3 MLD, thereby leaving a gap of 617.6 MLD sewage that remains untreated. This gap is ever expanding because urban populations tend to grow at a very fast pace.

There are 36 sewage treatment plants at 21 locations spread over a network of approximately 7000 km of sewage lines all across the Delhi region, comprising of all of the approved 44 re-settlement colonies and 126 urban villages, 541 unauthorized–regularized colonies and 100 unauthorized colonies (DJB, GNCTD 2017). Out of the 30 STPs studied by the CPCB (2005a, b) for their efficiencies, 11 failed to fulfill the general standards set for discharge in surface waters with the levels of biological oxygen demand (BOD) greater than 20 mg/l, 23 STPs failed the standards with total suspended solids (TSS) more than 30 mg/l and 12 STPs failed the standards with chemical oxygen demand (COD) higher than 100 mg/l. As per the observations made, less than 50% of the entire sewage of Delhi is actually being collected and treated. High BOD and fecal contamination have been the major setbacks even after treatment of the sewage.

Activated sludge process (ASP) is the most commonly employed technology in Class I cities of the country, which comprises of 59.5% of the total installed capacity. Up-flow Anaerobic Sludge Blanket (UASB) technology is the second most widely used, constituting 26% of total installed capacity, and is common among Class II towns. These are the two main technologies often used in combination with other primary or tertiary treatment units. The use of a series of waste stabilization ponds (WSP) is employed in 28% of the plants, though the combined capacity comprises only 5.6% (CPCB 2005a, b).

The ASP being the most commonly used process is employed at 23 STPs in Delhi. The other processes used for treatment include WSP, being used in one STP, trickling filters in one, High Rate Biofilters in 2, High Rate ASP + Biofilter in one and ASP–Extended Aeration in 2 of the STPs. STPs with ASP and trickling filter technologies are being used in combination with the primary sedimentation process (CPCB 2005a, b). All of the treated sewage is discharged into river Yamuna. Treated wastewater is also being used for irrigating crop fields in Delhi. Vegetables in Delhi cultivated over 1700 ha of land area are irrigated with wastewater near to Keshopur and Okhla STPs (Kaur et al. 2012). Each of the schemes used for treatment have their own pros and cons, but it is important to look into some of the aspects in order to assure their suitability before such applications or uses which have direct implications on human health.

Proper operation and maintenance of treatment plants as well as sewage pumping stations have largely been neglected all across the country. Around 39% of the existing plants do not conform to the EPA prescribed standards ensuring suitability for discharge into streams (CPCB 2005a, b). Inappropriate design, poor maintenance, frequent electric break downs and lack of technical man power lead to improper functioning of the facilities built to treat wastewater, and the plants remain shut most of the time (CPCB 2007).

WSPs are low-cost technologies used for the removal of organic matter, BOD and pathogens. However, they demand large surface areas and intricate designing. The treated effluent contains nutrients (nitrogen, phosphorus, etc.) and is therefore unsuitable for being discharged directly into surface waters. Moreover, the performance of WSPs is substantially lower in terms of their potential for meeting the TSS, BOD and COD discharge standards, and extremely poor in total/fecal coliform removal (Tare 2010). The potential of WSPs in generation of economically viable resources, such as fuel, manure, electricity and food, is extremely low.

Trickling filters are capable of treating biodegradable effluent including domestic black water/brown water/gray water. However, apart from primary sedimentation which is compulsory to avoid clogging of the filter beds, a secondary clarification step as well as post-treatment of excessive sludge is also mandatory for this method (WSP 2008). Moreover, trickling filters have specific requirements, such as continuous supply of power and wastewater, skilled design and maintenance. Special adaptations for cold climates are needed, such as proper insulation, lesser effluent recirculation and better distribution techniques (UNEP 2004).

Though the techniques used may be considered effective in the removal of BOD, COD, dissolved and suspended organics, volatile organic materials, nitrates, phosphates, etc., tertiary treatment processes may be useful in substantial removal of TDS. As against the secondary treatment processes that primarily remove organics to a large extent, the tertiary processes such as adsorption, electrolysis may be employed for the removal of toxic metals which are a major concern in river Yamuna.

Adsorption is a rapidly advancing process in this direction. It is believed to be one of the best and a universally used techniques that has an edge over other tertiary treatment methods available, for example, distillation, crystallization, evaporation, electrolysis, electrodialysis, solvent extraction, oxidation, precipitation, ion exchange reverse osmosis, micro- and ultra-filtration. Adsorption is a process which is rapid, low in cost and easy in maintenance and operation. Adsorption can also remove suspended solids, which no other tertiary process mentioned is capable of, except evaporation. The process is capable of efficient removal of solid and suspended materials, organic, inorganic and biological contaminants and finds application in reclamation, treatment as well as source reduction (Gupta et al. 2012). Volatile compounds are the only components that cannot be removed substantially by adsorption, but only solvent extraction method can, which lacks in the other merits of adsorption.

Recently, nanomaterials, such as carbon nanotubes and composites of carbon nanotubes, have been utilized for the efficient removal of heavy metals and organic pollutants (Gupta et al. 2012; Parween et al. 2014, 2016; Saleh and Gupta 2012). They are emerging as preferable adsorbents owing to their well-defined structure, large surface area, high aspect ratios, hydrophobicity and modifiable surfaces.

Wastewaters are highly polluted and may contain colored substances, solid waste containing organic, inorganic as well as biological pollutants. Colored substances have been treated in water/wastewater by several researchers as an important class of contaminants with regard to their potential for inducing carcinogenicity (Gupta et al. 2013; Saravanan et al. 2013, 2015a, b).

This calls for a need to employ a good combination of primary, secondary and tertiary treatment technologies. Wastewater in Delhi therefore demands a schematic process including treatment at primary, secondary as well as tertiary levels.

In this regard, a detailed investigation of diverse classes of pollutants in river Yamuna within the National Capital Territory of Delhi has been carried out. This would provide a baseline and help to identify the thrust areas that need immediate attention and seek measures that can improve execution and efficacies within the waste water treatment sector.

For this purpose, surface water samples from river Yamuna were collected in February 2013 from 14 locations

in the National Capital Region of Delhi and characterized for their quality with regard to nutrients, organic and inorganic pollutants.

Materials and methods

Study area and sampling locations

A detailed description of the study area can be found in the study conducted by Parween et al. (2014). Water samples of river Yamuna were collected in 2013 from 14 locations (L1–L14) starting from village Jagatpur, Delhi to Rajpur Khadar in Noida, UP to cover the polluted stretch of the river in Delhi region. Selection of locations (Fig. 1) was based on the drainage system of the river. Most of them are in receipt of effluents from major drains and canals carrying industrial waste, sewage and agricultural discharges and thus predicted to be the most polluted locations in the study area. L1 and L2 which are rural agricultural locations in the village Jagatpur lie remote to urban Delhi and are relatively unpolluted. The water quality at these locations seemed to be fairly better as compared to those lying downstream to them. L3 and L4 are urban locations lying downwind to the Narela industrial area. L5 and L6 are urban industrial locations which receive discharges from the Najafgarh drain and the supplementary drain. These are located downwind to a number of industrial areas such as Alipur, Badli and Bawana. L7, L8 and L9 are urban agricultural locations. They are fed by sewage, drains and by-products of Rajghat power plant, which is also located upwind. These also lie downwind to a number of industrial areas scattered from north to west of the megacity. L10 lies downstream to many drains and the Indraprastha power plant, which also discharges the effluents from its ash pond into the river. This location is marked by high traffic in the busy roads surrounding it. L11 and L12 are located at the Okhla/Kalindi Bird Sanctuary surrounded by thickly forested area. These are urban industrial and residential locations receiving effluents from STPs, various drains and the Hindon canal. The diversion of water to Agra canal leaves the river concentrated with pollutants beyond these locations. L13 lie in a high traffic zone. L14 receive discharges from the Shahdara outfall drain and agricultural runoff from UP. These are urban residential locations. Apart from the major drains mentioned above, several

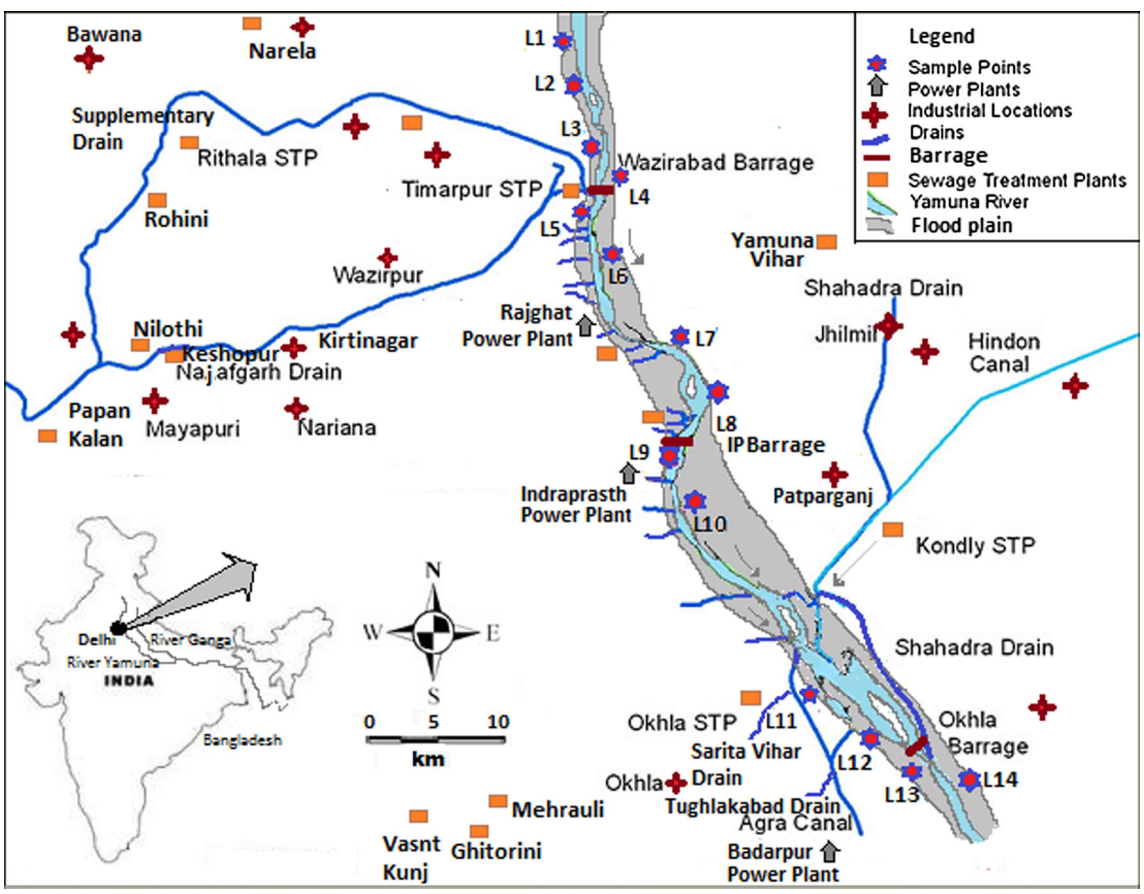

Fig. 1 Physiological map of Delhi region showing study locations along the floodplain of river Yamuna

other drains, effluents and agricultural discharges join the river upstream to various locations selected on either side of the river (Parween et al. 2014).

Sample collection and processing

Grab river water samples were collected in pre-cleaned polyethylene bottles. All the sample bottles were stored in iceboxes till brought to the laboratory for analysis. Samples were collected in duplicates. One set of samples was acidified with concentrated nitric acid (1.5 ml concentrated nitric acid per liter of the sample) and used for bulk metal analysis. The other set (unpreserved) was used for the analysis of pH, EC and ORP. The unpreserved samples were filtered before quantification of major ions. All the bottles were stored in a refrigerator at approximately 4 °C till analysis.

Extraction and analysis

Acidified samples were analyzed for heavy metals aluminum (Al), cadmium (Cd), chromium (Cr), copper (Cu), iron (Fe), manganese (Mn), nickel (Ni), lead (Pb) and zinc (Zn). Standard solutions for calibration of each metal were prepared by appropriate dilution of the stock solutions of 1000 ppm (Merck, Germany). Metals were analyzed using flame atomic absorption spectrophotometer (Thermo Scientific, M series). Instrument parameters for metal analysis were preset before analysis as per Grasshoff et al. (1999). Air/acetylene mixture was used as fuel. For each metal, a multi-point calibration graph was prepared. After examination of every 10 samples, blank and control standards were analyzed to examine the instrument and minimize errors. Anions, except phosphate (PO_4^{3-}), were analyzed with ion chromatogram equipped with 732 IC detector (Metrohm, Switzerland). PO_4^{3-} was determined by the ascorbic acid method (APHA 2005) using a spectrophotometer (HACH® Germany) at a wavelength of 650 nm. Bicarbonate analysis was conducted with the help of 877 Titrino Plus Autotitrator (Metrohm, Switzerland).

Quality assurance

The instruments were calibrated for each element before analysis. The analysis was performed only when the R^2 was greater than 0.995 with five-point calibrations. A blank and known standard were analyzed after every 10 analyses. Samples were analyzed in duplicates to ensure repeatability of the results. Samples were diluted accordingly for the analysis of each element so that the results were included well within the range of instrument detection limits. The lower detection limits ranged from 0.005 to 0.05 mg L^{-1}. The replicate analysis had relative standard deviations less than 5%.

Results and discussion

Water quality

Hydro-chemical characterization

Flowing water acquires a chemical composition which is reflected by the composition of the lithology it flows through. "Hydro-chemical facies" are used to describe the natural waters with varying chemical composition. The facies are a function of the lithology, solution kinetics and flow patterns of the aquifer. Hydro-chemical facies are classified on the basis of dominant ions in the water samples using Piper's tri-linear diagram. The concentrations of major ionic constituents of river water samples were plotted as a Piper tri-linear diagram (Piper 1953) to determine the water type (Fig. 2). The classification for cation and anion facies, in terms of major-ion percentages and water types, has been based on this diagram (Back 1966). The cationic field of the diagram identified the river water samples as the no dominant type. The predominant anion type of the water samples was bicarbonate, chloride and the no dominant type.

The diamond-shaped field represents the composition of water with respect to both cations and anions. The plot of data on the diamond-shaped tri-linear diagram (Fig. 2) indicates that the water samples fall under fields 1, 4 and 5. This indicated that alkaline earth metals exceeded the alkalis, strong acids were more than the weak acids, and the ions representing carbonate hardness (secondary alkalinity) exceeded 50%, respectively, with the fields. The data plots indicated the dominance of alkaline earth metals in the determination of the hydrochemistry of the region. The role of both strong and weak acids was also apparent, though dominated by the strong acids.

Water quality has been classified (Davis and De Wiest 1966) on the basis of total dissolved solids (TDS). Water with TDS up to 500 mg l^{-1} is desirable for drinking; 500–1000 mg l^{-1} is permissible for drinking; up to 3000 mg l^{-1} is useful for agricultural purposes; and above 3000 mg l^{-1} is unfit for drinking and irrigation. According to this classification, the river water was identified as permissible for drinking. Based on the electrical conductivity values, Richards (1954) classified irrigation water into four groups in order to assess salinity hazard (μS cm^{-1}); excellent (up to 250), good (250–750), fair/medium (750–250) and poor or bad (>2500). Excess salinity reduces the osmotic activity of plants and interferes with the plant absorption of water and nutrients from the

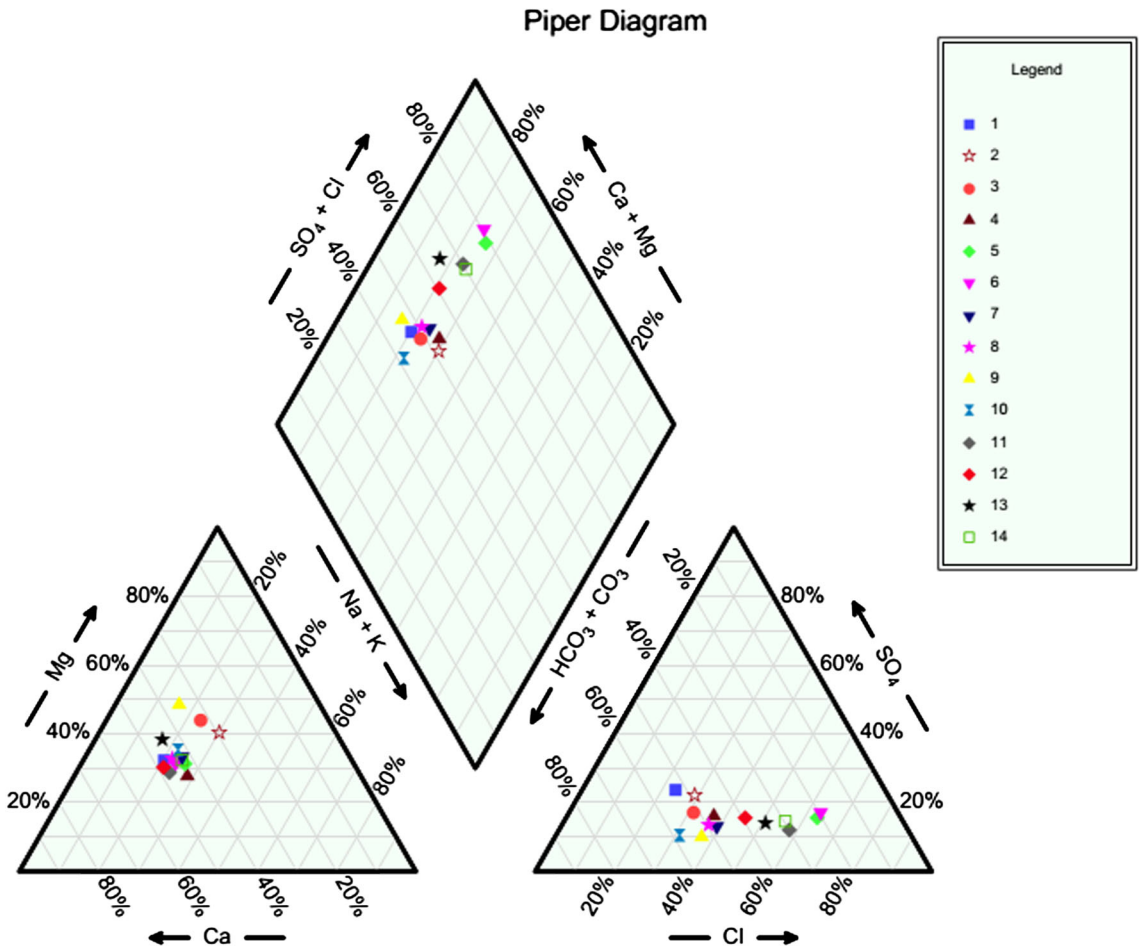

Fig. 2 Piper diagram for major ion distribution in river Yamuna, Delhi region

soil (Saleh et al. 1999). As per Richards's classification, the water samples are good to medium in salinity.

The concentration of sodium (Na) is important in classifying irrigation water. Excessive Na has undesirable effects that may change soil properties and reduce soil permeability. Natural waters are assessed for their suitability for use in agriculture in terms of percent of Na content (Wilcox 1948). Na in combination with carbonate can lead to the formation of alkaline soils, while Na along with chloride (Cl^-) forms saline soils. Both these soils are unfavorable for the growth of plants. The hydro-chemical quality of river Yamuna could be studied from plots between the percentage of Na and electrical conductivity on the Wilcox diagram.

Croplands irrigated with waters belonging to doubtful to unsuitable category decline in yield. This is probably from osmotic effects on plants due to the presence of excess Na salts. Sodium ions tend to be adsorbed by clay particles when present in excess, displacing magnesium (Mg) and calcium (Ca) ions. The exchange process reduces the permeability leading to poor internal drainage. Circulation of air and water is restricted during wet conditions, and such soils are usually hard when dry (Saleh et al. 1999). Wilcox diagram revealed that most of the samples fell under good to permissible category (Fig. 3).

The relative activity of sodium ion in the exchange reaction with soil is expressed in terms of a ratio known as sodium adsorption ratio (SAR). It is a measure of alkali/-sodium hazard for crops. The SAR values in the study area ranged from 0.5 to 1.1 with an average of 0.8. All samples had SAR <10 and fall under excellent water category as per Richards (1954). The data plot on the U.S. Salinity Laboratory Diagram (Richards 1954) illustrates that the samples fall into categories C2S1 and C3S1 (medium to high salinity with low sodium), indicating high salinity and low-sodium water type, suitable for irrigating most of the soil types (Fig. 4).

Ions and heavy metals

Major ions and heavy metals (mg l^{-1}) were found at a range of 20.7–64.8 (Na), 6–18.6 (potassium or K),

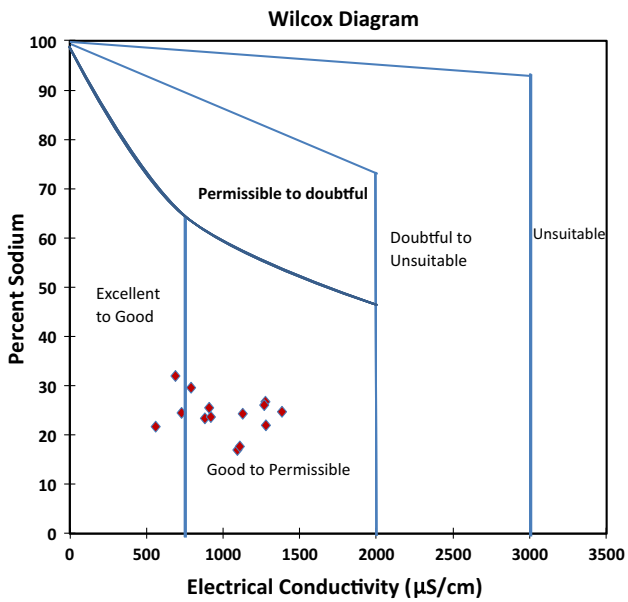

Fig. 3 Wilcox diagram for major ion distribution in river Yamuna, Delhi region

38.6–123.9 (Ca), 22.7–57.8 (Mg), 0.3–0.7 (fluoride or F^-), 47.7–293 (chloride or Cl^-), 0.9–3.9 (nitrate or NO_3^-), 38.9–99.3 (sulfate or SO_4^{2-}), 172–282 (bicarbonate or HCO_3^-), 0.2–1.3 (phosphate or PO_4^-), 2.5–8.2 (Fe), 0.4–0.9 (Al), 0.1–0.3 (Cu), 0.1–0.4 (Cr), 0.03–0.09 (Co), 0.02–0.09 (Cd), 0.2–0.8 (Pb), 0.2–0.7 (Mn), 0.2–0.5 (Ni) and 0.2–0.5 (Zn). The values corresponding to each location are enlisted in Table 1. Major ions are supplied to rivers via chemical weathering of the lithology of the basin and anthropogenic sources. Ca is the major cation in the study area. Yamuna river system (YRS) catchment consists of both limestones and dolomites (Singh et al. 1998) which are dominant sources of Ca. Together with Mg, it comprises of ~75% of the total cation (Tz^+) load in river water. Mg was correlated with Ca (Table 2). Both were correlated with HCO_3^-, an indication of carbonate weathering as the major source. Low Ca/Mg ratio (0.7–1.7) may be due to the supply of Ca and Mg via dissolution of dolomites that constitute a major portion of the Precambrian carbonates (Singh et al. 1998). Low Ca/Mg can also result if Ca is preferentially removed from the rivers by precipitation as calcite. An indication of greater Ca precipitation has been observed by Ravindra and Kaushik (2003) in the stretch of the river passing by Haryana (just before the river enters Delhi). It could either be of detrital origin or due to calcite precipitation or both. Major fractions of Na and K in the YRS are derived from silicate weathering. The K/Na ratio was quite low (0.1–0.3) in the Delhi stretch, which was lower as compared to the waters overhead with an average value of 0.6 (Dalai et al. 2002). The ratio has been observed to decrease in the lower reaches of the river, probably due to increasing Na concentration (expected from anthropogenic sources).

Bicarbonate is the dominant anion in the river water followed by Cl^-. Both were significantly correlated with TDS and constituted 84.5% of the total anion (Tz^-). The main sources of HCO_3^- in river water are carbonate and silicate weathering. Carbonate weathering seems to be the major contributor as evident from the correlation of HCO_3^- with Ca and Mg. The ratio (Ca + Mg)/HCO_3^- in the study area corresponded to 1.7 as against 1.33 in river Ganga in the Himalayan region (Chakrapani 2005). The ratio (Ca + Mg)/(HCO_3^- + SO_4^{2-}) was 1.3 indicating SO_4^{2-} to be another important and major anion. The river had $SO_4^{2-} > Ca$, possible due to the supply of SO_4 by pyrite oxidation.

The Na/Cl ratios in rains/snow have been used to assess the contribution of Na to rivers/streams from atmospheric deposition. This ratio shows a wide range with a typical mean value of 0.5 (Dalai et al. 2002). It is interesting to note that the average ratio was ~0.7 in the upper stretch of the river before Wazirabad barrage, but ~0.43 downstream to the barrage. This denotes a considerable excess of Cl^- over Na in the lower segment of the river. This indicated that there are other dominant sources of Cl^- apart from recycled marine salt and evaporite dissolution, likely anthropogenic. Studies with stable isotopes in YRS and their comparison with data on rainwater from New Delhi during the same period showed that the effect of evaporation in these rivers, if any, is minimal (Dalai et al. 2002). As opposed to the decreasing molar abundance in the headwaters of Yamuna: $HCO_3^- + CO_3^{2-} > SO_4^{2-} > Cl^- > NO_3^- > F^-$ (Dalai et al. 2002), the Delhi stretch had anion abundance in the order $HCO_3^- + CO_3^{2-} > Cl^- > SO_4^{2-} > NO_3^- > F^-$. Tremendous increase in the concentration of Cl^- is thus indicative of intense pollution loads due to anthropogenic activities. Chloride content exceeded the BIS (2012) guideline values for acceptable levels in drinking water at some locations. Excessive Cl^- in fresh water is considered as an index of pollution. Cl^- is found at very high levels in industrial effluents and sewage. Such high levels of Cl^- have also been observed by Ravindra and Kaushik (2003) in Haryana region.

The main sources of Ca comprise of weathering of rocks, seepage, drainage, wastewater, etc. It was high and exceeded the acceptable limit as per the BIS standard at many locations, particularly at locations L5, L6, L11, L12, L13 and L14. An increase in Ca concentration was observed downstream the river. The reason could be input from anthropogenic sources. The levels of Ca, at many locations, exceeded the BIS standard of 75 mg l^{-1} recommended to be acceptable for drinking water. Calcium, however, finds a mixed source in this study. Sources like agricultural application, sewage disposal and industrial

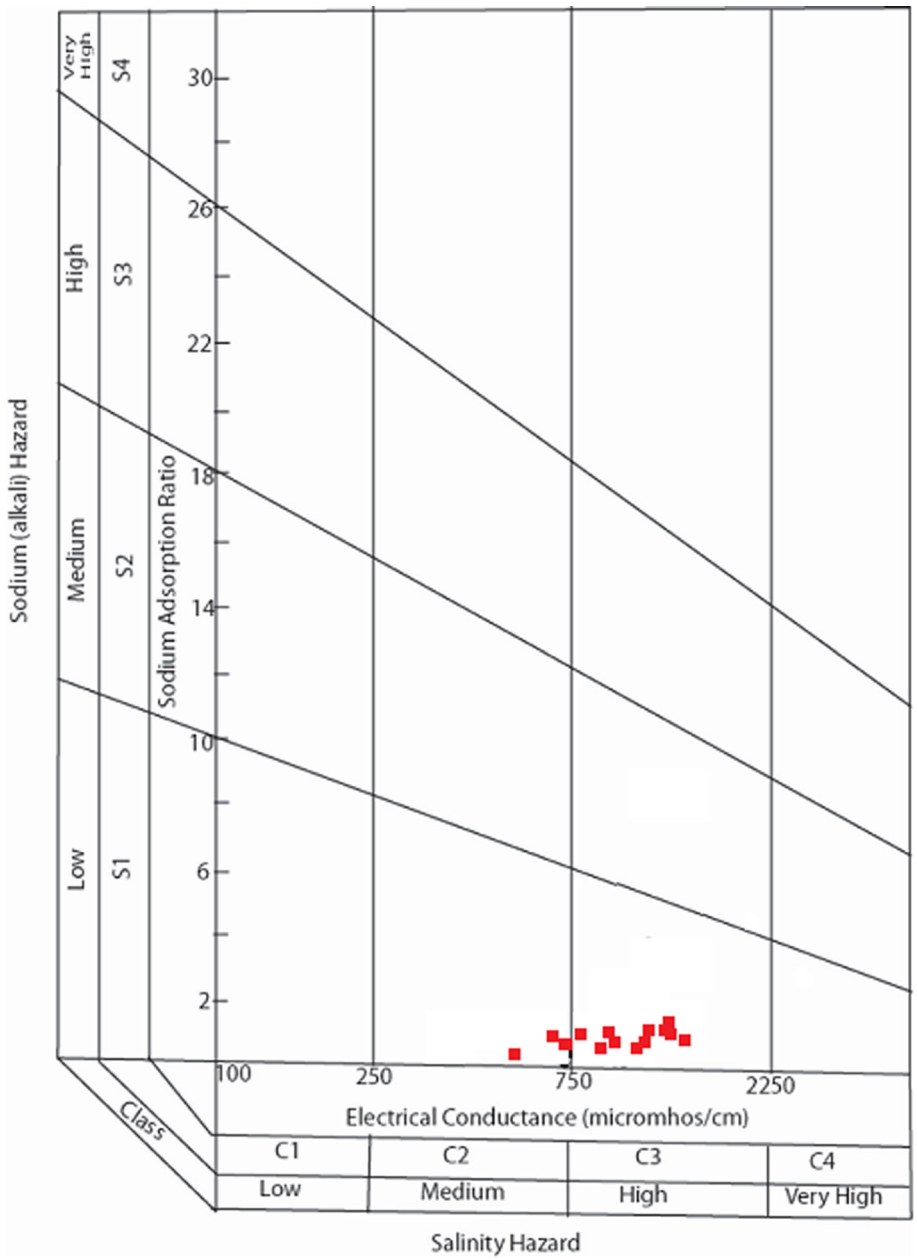

Fig. 4 US Salinity Laboratory Diagram for salinity and sodium hazards in river Yamuna, Delhi

effluents may have led to such high concentrations, as peak levels were observed immediately downstream to the discharge points of major drains into river Yamuna. L5 and L6 lie immediately downstream of the discharge point of the Najafgarh drain, the largest drain in Delhi, as well as the supplementary drain along with the return flow of the western Yamuna canal. L11, L12 and L13 lie downstream to the discharge point of Hindon canal, while L14 was fed by the Shahdara drain. The strong correlation of Ca with Mg, Cd, Zn, Cu, Cd, Cl^-, PO_4^{3-} and SO_4^{2-} (Table 2) indicated agriculture as a common source of these contaminants. All the three inlets to the river having such high levels of Ca carry discharges from agricultural activities in the western and eastern parts of Delhi and Uttar Pradesh. Moreover, the floodplain of the megacity itself is extensively used for agriculture throughout its stretch. L9 and L13 showed high levels of Mg. Not much variation was observed in the distribution for Mg. However, it exceeded the standard levels of drinking water (BIS 2012) at almost all the locations. Mg may have sources similar to that of Ca. Na showed a trend somewhat similar to that of Ca with concentrations higher at L5, L6 and L11 and downstream. Both were found to be correlated ($r = 0.68$, $p < 0.01$), probably due to origin from common sources, such as weathering of rocks, sewage and waste water. The pattern of K distribution was different and had little variation along

Table 1 Major ions and metals in river Yamuna, Delhi region

Location	Ions and trace metal concentration in river water of Yamuna, Delhi (ppm)																							
	Na	K	Ca	Mg	Fe	*Al	*Cu	*Cr	Co	*Cd	*Pb	*Mn	*Ni	*Zn	*F^-	Cl^-	NO_3^-	SO_4^{2-}	HCO_3^-	PO_4^{3-}	pH	EC	TDS	ORP
1	20.7	11.6	54.6	22.7	2.5	473	83	100	34	11	202	228	254	207	336	47.7	2.6	42.4	161.6	0.6	8.1	557	388	240.0
2	38.4	13.2	38.6	32.4	2.9	564	121	80	25	37	306	242	168	212	308	57.4	2.9	30.3	186.4	0.2	8.1	686	446	238.0
3	35.2	6.0	45.2	37.3	3.4	523	93	70	40	53	316	286	192	247	448	95.9	1.6	56.3	206.4	0.2	7.4	726	458	189.5
4	43.2	9.0	63.9	24.8	4.1	673	175	100	31	31	405	396	203	288	484	115.0	2.4	61.9	217.3	0.2	7.5	788	519	157.0
5	64.8	8.7	97.9	44.1	6.6	831	198	160	46	58	660	429	364	421	708	246.9	1.2	87.7	232.9	0.5	6.6	1273	775	94.7
6	50.7	7.7	90.2	38.4	8.2	878	245	280	76	74	837	644	451	473	624	216.8	1.0	74.5	213.2	0.9	6.3	1125	621	61.0
7	42.9	11.9	74.4	35.4	5.4	684	123	320	94	64	580	396	547	406	258	153.8	2.1	53.1	209.3	1.3	7.9	906	584	182.2
8	33.2	12.8	70.9	31.3	4.3	759	157	210	52	55	522	227	264	413	578	120.8	2.4	46.1	194.8	0.7	7.3	878	604	235.5
9	28.7	14.4	69.9	57.8	6.1	522	112	150	37	39	720	728	421	384	295	141.5	0.9	38.9	257.3	0.6	7.0	1090	695	45.5
10	37.8	16.0	76.5	38.9	6.0	465	91	140	45	55	498	571	272	278	441	116.6	2.1	43.4	272.2	0.2	7.8	917	743	69.6
11	55.3	18.6	116.0	41.7	4.5	716	228	480	97	93	610	333	351	395	475	276.7	3.5	70.4	233.4	1.2	6.9	1381	953	260.9
12	52.9	16.5	123.9	47.2	5.6	570	316	370	95	85	672	428	426	421	525	293.0	3.9	99.3	249.6	0.7	7.1	1276	782	247.0
13	40.6	12.9	108.5	56.4	4.4	652	139	290	74	65	653	232	424	401	400	235.2	2.5	71.0	248.2	0.8	6.8	1106	609	224.0
14	55.2	15.2	94.8	43.1	6.3	424	223	530	112	56	510	373	473	437	620	236.2	3.2	79.9	227.2	0.8	7.0	1265	712	249.0
BIS 2012			75	30	0.3	30	50	50		3	10	100	20	5000	1000	250	45	200			6.5-8.5		500	
WHO 2008	200	200	200	150	0.3		2000	50		3	10	400	70		1500		50	500						

* Concentrations in ppb; EC in μS/cm, ORP in mV, TDS in mg/l

Table 2 Correlation matrix for ions and metals in river Yamuna, Delhi region

	Na	K	Ca	Mg	Fe	Al	Cu	Cr	Co	Cd	Pb	Mn	Ni	Zn	F^-	Cl^-	NO_3^-	SO_4^{2-}	HCO_3^-	PO_4^{3-}	pH	EC (μS/cm)	TDS (mg/l)	ORP (mV)
Na	1.0																							
K	0.1	1.0																						
Ca	0.7	0.5	1.0																					
Mg	0.3	0.3	0.6	1.0																				
Fe	0.6	0.0	0.5	0.5	1.0																			
Al	0.5	−0.4	0.3	0.0	0.4	1.0																		
Cu	0.8	0.2	0.8	0.2	0.5	0.4	1.0																	
Cr	0.6	0.6	0.8	0.4	0.4	0.1	0.7	1.0																
Co	0.6	0.4	0.8	0.4	0.4	0.1	0.6	1.0	1.0															
Cd	0.7	0.3	0.8	0.5	0.5	0.4	0.7	0.7	0.7	1.0														
Pb	0.5	0.2	0.7	0.7	0.8	0.5	0.6	0.5	0.5	0.7	1.0													
Mn	0.1	0.0	0.2	0.4	0.8	0.1	0.2	0.0	0.0	0.2	0.6	1.0												
Ni	0.4	0.2	0.6	0.6	0.7	0.2	0.4	0.7	0.8	0.5	0.7	0.4	1.0											
Zn	0.6	0.2	0.8	0.5	0.8	0.5	0.7	0.7	0.7	0.7	0.9	0.4	0.8	1.0										
F^-	0.7	−0.2	0.5	0.0	0.5	0.4	0.6	0.3	0.3	0.4	0.4	0.1	0.1	0.5	1.0									
Cl^-	0.8	0.3	1.0	0.6	0.6	0.4	0.8	0.8	0.8	0.8	0.8	0.2	0.7	0.8	0.5	1.0								
NO_3^-	0.2	0.6	0.3	−0.1	−0.4	−0.3	0.4	0.5	0.5	0.3	−0.3	−0.5	0.0	−0.1	0.0	0.3	1.0							
SO_4^{2-}	0.8	0.0	0.8	0.4	0.5	0.3	0.8	0.6	0.6	0.6	0.5	0.1	0.5	0.7	0.7	0.9	0.3	1.0						
HCO_3^-	0.4	0.4	0.6	0.8	0.6	−0.1	0.3	0.3	0.3	0.5	0.6	0.6	0.4	0.4	0.2	0.6	−0.1	0.4	1.0					
PO_4^{3-}	0.2	0.3	0.6	0.2	0.3	0.3	0.3	0.7	0.8	0.6	0.5	0.0	0.8	0.7	0.0	0.5	0.2	0.3	0.0	1.0				
pH	−0.6	0.1	−0.7	−0.6	−0.7	−0.6	−0.6	−0.4	−0.4	−0.6	−0.8	−0.4	−0.5	−0.8	−0.7	−0.8	0.3	−0.7	−0.4	−0.3	1.0			
EC	0.8	0.4	0.9	0.7	0.7	0.3	0.7	0.8	0.7	0.8	0.8	0.4	0.7	0.8	0.5	1.0	0.1	0.7	0.7	0.5	−0.8	1.0		
TDS	0.7	0.6	0.8	0.6	0.6	0.2	0.6	0.7	0.6	0.8	0.7	0.4	0.5	0.6	0.4	0.8	0.2	0.5	0.7	0.4	−0.6	0.9	1.0	
ORP	0.0	0.3	0.1	−0.2	−0.6	−0.2	0.2	0.4	0.3	0.1	−0.4	−0.8	−0.1	−0.1	−0.1	0.1	0.8	0.1	−0.4	0.2	0.3	−0.1	−0.1	1.0

Two-tailed Pearson's correlation coefficients; values > 0.532 are significant at $p < 0.05$; values > 0.661 are significant at $p < 0.01$ ($n = 14$)

the different locations. It was highest at L11 followed by L12 and L10. Agricultural practices seem to influence the concentration of K in river Yamuna. This can be supported by its correlation with NO_3^- ($r = 0.64$, $p < 0.01$), both commonly used in fertilizers. L1 and L2 had K concentrations higher than those lying downstream to them. This is probably due to the contribution of K from the agricultural fields in Haryana. A rise in the level was seen from L7 to L14, probably again due to the increased agricultural practices in the region. Cl^- concentration was too high at L5, L11, L12, L13 and L14. Such high levels of Cl^- could probably be from sewage contamination or agricultural input. This strong correlation (Table 2) of Cl^- with Na^+ and Ca^{2+} can be attributed to sewage and industrial effluents from the drains joining the river (Trisal et al. 2008). A high proportion of Ca^{2+}, Mg^{2+}, SO_4^{2-} and Cl^- can be indicative of agricultural influence (MFE 2008). The correlation of Ca^{2+} with Mg^{2+}, SO_4^{2-}, Cl^- and PO_4^{3-} (Table 2) is indicative of their common agricultural origin. Therefore, Cl^-, Na^+, Ca^{2+} and Mg^{2+} seemed to have diverse sources. L1 was marked by higher levels of NO_3^- and PO_4^{3-} as compared to locations downstream, probably due to intensive agricultural activities in Haryana which lie upstream to it. However, there was a considerable rise in concentration downstream the river. Both were high at L11 and subsequent downstream locations. The Hindon cut which joins river Yamuna upstream to L11 carrying the agricultural discharges and sewage may be responsible for such high concentrations. The river is highly eutrophic with algal blooms visible at many regions. Though NO_3^- and PO_4^{3-} seemed to share a common source, there still was variation in the distribution of both. Agriculture seemed to be the main source of NO_3^- as against PO_4^{3-}, which was probably from agriculture, as well as sewage and industrial discharges. The levels of PO_4^{3-} peaked at L7, which carries effluents from the Najafgarh drain. The drain is fed by several industrial effluents as well as STPs. Fluoride has been identified as a naturally occurring pollutant in groundwater within several parts of the country. However, the correlation of F^- with SO_4^{2-} indicated the use of fertilizers as a common source. The common anthropogenic sources of F^- in the environment are coal combustion, waste from numerous industries, including steel, aluminum, copper and nickel smelting, and the production of glass, phosphate fertilizers, bricks and tiles (Skjelkvsle 1994). SO_4^{2-} too is derived from a number of anthropogenic sources such as fertilizers, industrial discharge, sewage, coal combustion. (MFE 2008). High levels of SO_4^{2-} were detected at L5 and L6 which indicated industrial discharges being a common contributor, while high concentrations at L1 and L2 are indicative of agricultural source at the locations. A rise in the concentration of at L14 could again be due to both industrial discharges from Shahdara drain as well as agricultural sources. SO_4^{2-} showed a concentration highest at L12, but F^- did not. This may probably be due to inputs from STPs (Kondly, Okhla, etc.) upstream to the location. Therefore, sulfate finds a mixed source in the region. Major cations, HCO_3^- and Cl^-, were correlated with TDS, being the dominant components of river Yamuna.

River water was contaminated with high concentrations of heavy metals in the study area. A comparison of the level heavy metal in the surface water of river Yamuna with those of the urban areas of other rivers has been provided in Table 3. Most of the heavy metals increased in concentration toward downstream of the river, with concentrations higher at locations L5, L6, L7, L9, L11 and L12. The impact of urbanization and industrialization was evident from the pattern of heavy metal distribution in river water. The locations downstream to the discharge point of Najafgarh drain (L6) and the one near Okhla industrial area (L12) could be considered as hotspots in terms of contamination with heavy metals. The basic character of the river water, as well as high carbonate and illite contents in the sediments, may be the reason for the high heavy metal accumulation (Singh 2001).

Fe was manifold higher in concentration at all the locations as compared to the WHO (2008) and BIS (2012) standards set for drinking water (Table 1). Fe was highest at L6, followed by L5 and L14. High levels of Al were found at L6 and L5. Values at all the locations exceeded manifold the WHO and BIS standards. L5 and L6 are located downwind to various industrial areas and may act as receptor sites for industrial emissions via atmospheric deposition. They also receive effluents from Badli, Bawana, Wazirpur and Najafgarh Industrial areas via the Najafgarh drain. Iron and steel processing, electroplating and metal processing, paper mill and chemical industries located in these industrial areas (Rawat et al. 2003) may be responsible for high Fe and Al. Fe visibly reached the river mostly from drains carrying industrial effluents as it was higher at L9, L10 and L14. Whereas L9 and L10 are fed by a number of drains, L14 receives discharges from the Shahdara drain, the second largest in Delhi. This can be supported by the fact that Fe is correlated with Pb, Mn, Ni, Zn and Cl^- (Table 2), all of which have origin from similar industries, especially electroplating and anodizing industries.

Fe measured in river water is Fe^{2+}, the reduced state of iron. Fe^{3+} precipitates in the form of minerals, such as carbonate, and is therefore not expected to be present in dissolved form. Fe is not considered as a contaminant even at high concentrations because of its not toxic nature for humans. The presence of Fe^{2+} in water samples is indicative of anoxic nature of the aquifer at least in some zones. Natural water bodies are generally stratified, having

Table 3 Comparison of heavy metal contamination (mg/l) of the study area with those of the other tropical rives in the continent

Location	River	Cd	Cr	Ni	Pb	Zn	Cu	Mn	Reference
Industrial city area of river in China	Yellow	–	#0.4–4.3	–	0.03–0.05	0.15–0.23	0.04–0.21	–	Fan et al. (2008)
Peripheral river water around Dhaka City (surface water discharge)	Buriganga	0–0.03	0–0.27	0.007–0.15	0.02–0.47	0.24–4.6	–	–	Khan and Biswas (2013)
	Balu	0.001–0.08	0.002–0.73	0.008–0.35	0.003–0.34	0.11–3.0			
	Torag	0–0.07	0–0.11	0.007–0.1	0.006–0.39	0.17–2.05			
	Tongi khal	0.009–0.1	0.008–0.8	0–0.05	0.01–0.45	0.08–2.51			
	Shitalakhya	0–0.8	0.001–0.27	0.005–0.007	0.004–0.23	0.11–3.55			
Surface water around urban areas of Karachi, Pakistan	Malir	0.020–0.070	0.690–1.260	–	0.380–0.760	–	0.260–1.000	–	Sharmeen et al. (2014)
Surface water from urban area of Delhi	Yamuna	0.02–0.09	0.07–0.53	0.17–0.55	0.2–0.84	0.2–0.48	0.08–0.32	0.23–0.73	Present study

Concentrations in (μg/l), * mean values

an oxic layer above the middle zone of progressively declining DO and a complete anoxic layer at the bottom. River Yamuna is characterized by a low flow condition in Delhi owing to its meandering stage in the study area and can rather be considered to be a typical example of degraded lotic ecosystem reduced to sewage. The low values of ORP (Table 1) also confirm the miserable condition of the river in the stretch.

Al was correlated with Pd, Cu and Zn, which like Al are also known to originate from industries, especially electroplating industries. It was high at location L8, L11 and L7. The rise in levels at L11 may be due to the discharges from the Hindon canal which is fed with effluents from various industrial areas (Fig. 1). L8 and L7 lie downwind to the Rajghat power plant and may be in receipt of atmospheric deposition as well as ash pond discharges, thereby leading to high levels of Al produced from coal combustion.

Fe, Al, Pb and Zn were found to be negatively correlated with pH of the river water. Metal cations, including heavy metals and Al, typically increase in solubility with decreasing pH (Adriano 1986). The solubility of heavy metal contaminants depends on chemical speciation of the metal, the pH and redox potential of the surrounding. These determine the partitioning of metals between water-dissolved and particle-bound or soil-bound states. The solubility of heavy metals has been shown to be closely related to pH (Adriano 1986) and to changing redox potential (Chuang et al. 1996). With decreasing pH, the solubility of metal cations typically increases due to desorption from suspended soil mineral surfaces or organic matter. They are thus expected to occur preferentially in dissolved form. The solubility of minerals such as metal oxides, hydroxides and carbonates increases with decreasing pH. Fe, Mn, and Al are mostly present as oxides and hydroxides. They provide chemisorption sites to both cations and oxyanions. Therefore, the presence of hydrous metal oxides of Fe, Al and Mn can strongly influence dissolved metal concentrations because they can remove cations and oxyanions from solution by ion exchange, specific adsorption (chemisorption) and surface precipitation (Dzombak and Morel 1987). Most of the heavy metals are considered to be bound to oxide minerals by chemisorption via exchange of metal cations and oxyanions with surface protons or hydroxyls and form partial covalent bonds with the lattice ions (Alloway 1995). Manganese was correlated with Pb. Aluminum was found to be correlated with Pb and Zn. Iron correlated with Pb, Ni and Zn. The three abovementioned groups of correlated metals could be the result of the potential role of oxides and hydroxides of Fe, Mn and Al in adsorption of heavy metals Pb, Zn and Ni. Oxidation–reduction reactions control the transformation and reactivity of Fe and Mn oxides, which can adsorb heavy metals and metalloid pollutants and serve as major sinks for these pollutants (Huang and Germida 2002). At oxic–anoxic interfaces, the sediment–water interfaces, or within sediments, Fe and Mn may transform between the dissolved and solid forms. A decrease in dissolved Fe and Mn has

been observed at high redox potential (Husson 2013). This may be the reason why Fe and Mn were found to be negatively correlated with ORP of river water.

Cu and Cd had a concentration maximum at L12 followed by L6. Though Zn did not exhibit much variation, it was highest in L6 and L12. The source of these metals could be industrial effluent arising out of smelters or metallurgical processes, electroplating, paint and fabric printing industries (Lee and Duffeld 1977) joining the river upstream to L6 via the Najafgarh drain; while those joining the river before L12 reached by the Hindon canal. This can be supported by correlation of both Cu and Cd with metals which have industrial sources, namely Co, Pb, Zn and Cr (Rawat et al. 2003). Cadmium has a significant correlation with Cl^-, indicating the possible role of chloride ions in enhancing Cd solubility from soil to solution/water by formation of relatively stable chloride ion complexes, such as $CdCl^+$ and $CdCl_2$ (Weggler et al. 2004). Cl^- may thus augment the uptake of Cd (McLaughlin et al. 1994) by the aquatic organisms.

Cr was at its highest level in L12 followed by L7. Most of the locations downstream to Najafgarh and Okhla industrial area had high levels of Cr, thereby supporting industrial effluents as its major sources. The source of Cr may be the electroplating, paper printing and chemical industries predominant in Wazirabad, Badli and Mangolpuri areas. Rawat et al. (2003) reported high concentrations of Cr in the effluents from these industrial areas.

Co was found at high levels at L7 and L12. Both Co and Cr are used to make alloys and are expected to originate from metal processing and ceramic industries (ATSDR 2000). Concentrations of both Co and Ni were highest at location L7. High concentration of Fe, Cr and Cd could also be observed at location L7. All of these metals were correlated indicating atmospheric deposition from power plant emissions or Badli, Bawana and Wazirpur industrial areas as common sources. The location lies near Rajghat power plant, and all of these metals are also emitted from coal combustion.

Pb, Mn and Ni were at their maximum concentrations in L6, L9 and L14, respectively. L6 and L14 are receptors of industrial discharges from the Najafgarh and Shahdara drains. The three metals were high in concentration at L9 and have been identified as major products of smelters and metallurgical processes (Lee and Duffeld 1977). L9 is also under the direct influence of power plant emissions from Indraprastha power plant. Therefore, atmospheric deposition from coal combustion cannot be ruled out at this location. The correlation of Ni with Cu, Cd, Cr, Co, Fe, Pb and Zn was indicative of coal combustion as the major source of contamination (Klein et al. 1975). The high concentration of Mn at L6 may be related to the steel processing industries at Wazirpur, Badli and industrial areas. Pb and Cd levels have been found to be higher at locations downstream of the discharge point of major drains, probably from battery manufacturing units spread across the industrial areas (Rawat et al. 2003). Non-ferrous smelters too are a common source (Tahri et al. 2005).

The removal of pollutants from water at point sources holds utmost importance in megacities like Delhi, as it is a major sector that has immense potential of making a massive impact on the water quality of river Yamuna. An efficient treatment of waste water should involve three stages of treatment, namely primary, secondary and tertiary. Several methods have been employed for the purpose, mostly up to the secondary levels in the Delhi region followed by the discharge of effluent into river Yamuna. While treatment of only a fraction of the total wastewater generated is a major setback, it is also important to assess the suitability of the methods employed for the treatment process in close relation with the specific use aimed at post-treatment.

The observations of the study conducted emphasizes on the abundance of nutrients as well as toxic metals in the waters of river Yamuna, which are reasons of grave concern. These contaminants may undesirable for the health of water bodies as well as living organisms. Heavy metals which are toxic in nature and have been detected several folds higher than the limits prescribed by the BIS needs special attention. Such waters may be hazardous when used for drinking or irrigation of crop fields owing to their toxic levels as well as bioaccumulation through the food chain. Special mechanisms/processes need to be incorporated in the treatment of waste water for the efficient removal of toxic metals.

Conclusion

Hydro-chemical investigations were performed to assess the water type and quality. The predominant anion type of the water samples falls under bicarbonate, chloride and no dominant types. The data plots on the Piper diagram indicated the dominance of alkaline earth metals in the determination of the hydrochemistry of the region. The role of both strong and weak acids was also apparent, though dominated by the strong acids. The river water was identified as permissible for drinking with respect to the levels of EC. Richards's classification indicated the water samples to be good to medium in salinity. Based on the sodium content, the river water falls within the permissible category. Data plotted on the U. S. Salinity Laboratory Diagram illustrates the samples to be of high salinity and low-sodium water type, suitable for irrigating most of the soil types. Wilcox diagram rated the water quality within the permissible category.

Ca is the major cation in the study area. Bicarbonate is the dominant anion. Carbonate weathering is the major contributor of HCO_3^-, Ca and Mg. Sulfate was observed as an important major anion. High concentration of Cl^- indicated intense pollution loads due to anthropogenic activities. Chloride exceeded the BIS (2012) guideline levels for drinking water at some locations. Major ions and heavy metals were found to originate from anthropogenic activities as a result of rapid urbanization of the capital city. Agricultural, industrial and sewage discharges were recognized as the main anthropogenic sources of cation and anions.

All the locations downstream to the Wazirabad barrage were found to be contaminated with high levels of heavy metals in the river water. Heavy metals such as Fe, Al, Cu, Cr, Cd, Pb and Mn were found at concentrations beyond the standard levels for drinking water recommended by WHO and BIS. Low ORP values indicated reduced aquatic environment and high organic load in the river. The physicochemical parameters, especially pH and ORP, were important in determining the distribution of heavy metals in river water. Oxides and hydroxides of Al, Fe and Mn seemed to play role in retention of heavy metals in the river water. Industrial activities served as major contributors to contamination with heavy metals. In addition to it, agricultural activities along the floodplain were important sources of high metal concentration in the river water. Eutrophication was evident due to significant levels of NO^{3-} and total PO_4^{3-}, as well as lower ORP. Discharges from agriculture, sewage and power plants too played important roles in the rise of metal concentration. The Najafgarh drain and the Shahdara drain were the main carriers of an anthropogenic fraction of metals and ions to the river. The Hindon canal was also important in the contribution of metals in the river.

Water quality was poor at all of the locations with respect to heavy metal contamination, especially along the lower section of river Yamuna in the Delhi stretch. The heavy metals are the major contaminants arising mostly from the ever-growing industrial effluents and the coal-based power plants in the megacity. Anions such as chloride, phosphate and nitrate were also important contributors to poor water quality in the region. This emphasizes the need for special treatments for the prevention of metal contamination in the river, through both technologies as well as the implementation of strict regulations in order to sustain the exponentially growing population in megacities.

Waste water treatment is extremely important in terms of prevention of river water pollution form point sources. While less than 50% waste is being treated at present, 100% wastewater treatment is severely required in the megacity. Measures should be taken to ensure proper functioning, maintenance, uninterrupted power supply and regular monitoring of the treatment plants. Waste water treatment plants as well as sewage treatment plants in Delhi use various techniques that may be considered effective in the removal of BOD, COD, dissolved and suspended organics, volatile organic materials, etc. However, tertiary treatment processes are designed specifically to remove dissolved substances, nutrients and toxic metals. Tertiary treatment processes such as adsorption, electrolysis may be employed for the removal of toxic metals which are a major threat to the quality of river Yamuna in the Delhi region. A comprehensive treatment scheme need to be envisioned for substantial removal of soluble and insoluble pollutants, organic and inorganic contaminants, toxic elements and nutrients for safe reuse of treated wastewater.

While it is wise to reuse and recycle the treated waste in order to reduce the load on the limited resource of water, care should be taken to assure minimal risk to human health. Various measures should be adopted to ensure proper and efficient management of waste water. A list of recommendations follow to ensure the same.

Recommendations

(a) Parts of the sewage at some STPs are treated up to the secondary level and/or tertiary level, while the rest is left untreated. It is important to treat the complete amount of sewage that is produced in the city. Sewage should be treated systematically up to the tertiary level in the Delhi region before being discharged into rivers.

(b) A comprehensive treatment scheme need to be envisioned for substantial removal of soluble and insoluble pollutants, organic and inorganic contaminants, toxic elements and nutrients for safe reuse of treated wastewater.

(c) Conventionally treated sewage is not fit for irrigation. STP using treatment processes that retain significant nutritional value should be reused for irrigation post clarification/tertiary treatment. This would also reduce pollution of river body with nutrients. Sewage from STPs that are monitored regularly and provide high-quality effluent should only be allowed for use in agriculture.

(d) Waste should be recycled into useful products such as fertilizers, fillers, building material.

(e) Toxic wastes should be buried underground within sealed and leak-proof compartments rather than being incinerated.

(f) Waste water may be used in agriculture only if it is treated up to the tertiary level. However, it should not be used for the irrigation of crops that are eaten raw. Consumer awareness may be provided to promote

sufficient washing and cooking in order to reduce the load pathogens.

(g) Use of wastewater use in public park and gardens, golf course, tree plantation, city forest plantation, etc., should be promoted.

(h) Personnel who operate the plants must be imparted with adequate knowledge and training. Experts need be engaged in managing and monitoring with regular visits.

(i) Auxiliary power backup facility is recommended at all the STPs.

(j) Research and development on waste water treatment methods and processes should promoted so that low-cost, low-tech, user friendly methods may be developed and existing processes be refined for their efficiencies and effectiveness.

(k) The use of methods such as constructed wetlands that are low in cost, lesser energy intensive and easy to operate should be investigated as alternatives.

(l) Education and awareness among the farmers promoting adoption of modern methods in agriculture like drip irrigation, cropping systems, cultivation of remunerative non-edible crops and sewage water application methods.

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